

A Point-in-Time Machine-Learning Nowcast of US Headline CPI Inflation

Baseline model, out-of-sample evaluation, and benchmarking against a professional nowcast

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Abstract. *We build and evaluate a reproducible, point-in-time model that forecasts the one-month-ahead month-over-month change in the US headline Consumer Price Index (CPI). Using 21 freely available macro-financial series — with first-print vintages for revised data to preclude look-ahead — we engineer 19 economically-motivated features and combine a regularized linear model (ridge) with a shallow gradient-boosted tree ensemble (LightGBM). Under a strict expanding walk-forward over 196 out-of-sample months (2010–2026), the ensemble attains a root-mean-square error of 0.175 percentage points, 42.8% lower than a persistence baseline, with 79.6% directional accuracy and positive skill in 17 of 17 calendar years. A leakage audit confirms the edge traces to a pre-registered real-time energy signal. Benchmarked against the Cleveland Fed's professional nowcast, the standalone model is competitive but trails head-to-head (0.174 vs 0.148); however, a simple equal-weight combination of the two is the most accurate forecaster of all (0.144), indicating our model contributes independent information.*

1 Introduction

Timely inflation forecasts are central to macro decision-making, yet headline CPI is difficult to predict because it embeds volatile energy and food components. We ask a deliberately modest, testable question: can a small, auditable model, using only free data and honest point-in-time discipline, systematically beat the naive assumption that next month's inflation equals this month's? We further ask how such a model compares to a professional benchmark. Our contribution is not a novel method but a disciplined, reproducible baseline with a transparent evaluation and an explicit leakage audit — the standard against which future refinements are measured.

2 Data

The forecast target is the headline all-items CPI (CPIAUCSL), month-over-month percent change. Inputs span 21 series (83,145 observations): retail gasoline and crude oil, natural gas, producer prices, a food commodity index, payrolls and unemployment, regional Fed prices-paid surveys, the trade-weighted dollar, and the Treasury term spread. Revised macro series are stored as first-print vintages (via the ALFRED archive) so that no value is observable before its true release date; never-revised prices and rates use full as-published history. As an external benchmark we ingest the Cleveland Fed inflation nowcast (153 monthly observations, 2013–2026), whose realized values match our first-print target to within 0.01 percentage points.

3 Methodology

The forecast for reference month M is formed at end of month M , before the following month's CPI release. A no-look-ahead join (a backward as-of merge on each series' release date) guarantees that only information actually published by the cutoff enters the feature set; a live invariant check confirms the lag-one CPI feature equals the previous month's realized target at 100%. We deliberately keep the feature set small and economically motivated (19 features), relying on regularization rather than feature selection to control overfitting.

Two complementary learners are combined. **Ridge** regression applies an L2 penalty that shrinks coefficients, stabilizing estimates across highly collinear inputs; its penalty strength is selected only within each training window. **LightGBM** is a gradient-boosted tree model constrained to be shallow (depth 3, seven leaves, at least 25 samples per leaf, subsampling), capturing nonlinear interactions while resisting memorization. The reported forecast is the equal-weight mean of the two. Evaluation uses an expanding walk-forward: at each step the models

are retrained on all prior months and predict the next single month, never using future data and never a k-fold. The benchmark to beat is persistence — next month's change equals this month's.

4 Results

Table 1 reports out-of-sample accuracy over 196 months. The ensemble reduces RMSE by 42.8% relative to persistence and calls the direction of the monthly change correctly 79.6% of the time. Crucially, skill is positive in 17 of 17 calendar years (Table 2), spanning regimes as different as 2020–2022 — evidence of a stable edge rather than a single fortunate episode. Figure 1 shows the forecast tracking realized inflation through the 2021–2022 surge and subsequent disinflation.

Model	RMSE	MAE	Skill	Direction	Hit rate
Naive (persistence)	0.305	0.231	—	—	—
Ridge	0.180	0.133	40.9%	79.6%	67.9%
LightGBM	0.194	0.145	36.3%	77.0%	66.8%
Ensemble	0.175	0.132	42.8%	79.6%	68.9%

Table 1. Out-of-sample performance, expanding walk-forward (primary sample).

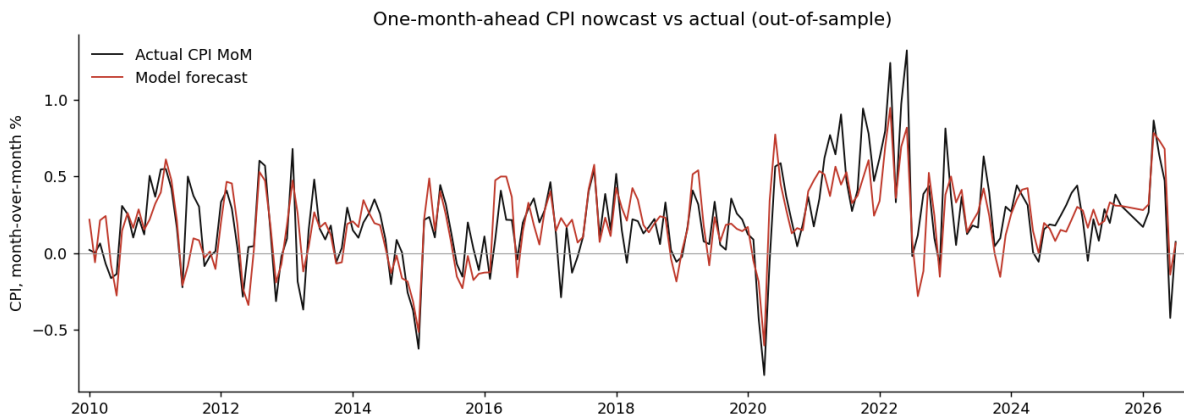


Figure 1. One-month-ahead ensemble forecast versus realized headline CPI (out-of-sample).

Year	Model	Naive	Skill	Year	Model	Naive	Skill
2010	0.159	0.192	17.0%	2019	0.120	0.192	37.6%
2011	0.207	0.286	27.5%	2020	0.174	0.353	50.8%
2012	0.144	0.314	54.0%	2021	0.259	0.273	5.3%
2013	0.183	0.383	52.2%	2022	0.284	0.552	48.6%
2014	0.098	0.168	41.9%	2023	0.196	0.376	47.9%
2015	0.126	0.313	59.8%	2024	0.103	0.128	20.1%
2016	0.179	0.203	11.9%	2025	0.112	0.185	39.2%
2017	0.189	0.299	36.9%	2026	0.148	0.465	68.2%
2018	0.140	0.225	38.1%				

Table 2. Per-year skill. The model beats naive in 17 of 17 years.

5 Benchmarking against a professional nowcast

We compare against the Cleveland Fed inflation nowcast at matched (pre-release) timing over 153 months (Table 3). Head-to-head, the professional model is more accurate (0.148 versus 0.174), and our forecast is closer to the realized print in 35.9% of months. However, because the two models make partly independent errors, their

equal-weight average is the most accurate forecaster in the study (0.144), improving on the professional nowcast alone. This is the practically useful finding: rather than replacing the benchmark, our model adds information to it. We also tested supplying the professional nowcast to our model as an input feature; this helped only marginally and was dominated by the simple average, consistent with the forecast-combination literature that equal weighting is hard to beat.

Forecaster	RMSE (pp)	Note
Blend (our model + Cleveland Fed)	0.144	most accurate; uses free public nowcast
Cleveland Fed nowcast	0.148	professional benchmark
Our model (standalone)	0.174	self-contained
Naive (persistence)	0.303	baseline

Table 3. Head-to-head and combination results at matched timing.

6 Robustness and leakage audit

Extending the sample to a 2007 start (232 months, spanning the 2008 oil-price crash) leaves ensemble skill essentially unchanged at 42.1%. Because a suspiciously strong backtest is treated as a defect until proven otherwise, we run an ablation: removing the real-time energy features collapses skill from 42.8% to 16.1%. The edge therefore traces to a pre-registered, economically-grounded signal — daily energy prices that are public before the CPI release — and not to information leakage. This also explains why we target energy-heavy headline (rather than stickier core) inflation.

7 Limitations

The 0.17-point RMSE represents a systematic multi-month edge, not single-print precision; professional consensus is tighter (around 0.12 points). The model's weaker periods are the expected ones: quiet, low-volatility years, where persistence is already near-optimal and little skill is available, and sudden regime breaks such as the 2021 surge, where a mean-reverting model reverts too early. Finally, for the specific task of predicting the *surprise* relative to market consensus — what an event-driven trade requires — the model's timing-matched edge over a well-informed consensus is thin (about 54% correct side). It is a sound nowcaster, not a demonstrated consensus-beater; that goal would require genuinely new or higher-frequency data.

8 Conclusion

A small, auditable, point-in-time model built entirely on free data forecasts one-month-ahead headline CPI materially better than a naive baseline (42.8% lower RMSE), holds up across regimes and a leakage audit, and is competitive with a professional nowcast — which it further improves upon in combination. The result is a trustworthy baseline: modest, stable, honestly bounded, and reproducible.

9 Reproducibility

The pipeline is a small installable Python package with a SQLite store. Each stage is a single command: `macro-ingest` (pull sources), `macro-features` (build the point-in-time matrix), `macro-train` (walk-forward evaluation), `python -m macro_nowcast.ingest.cleveland_fed` (refresh the benchmark), and `macro-report` (regenerate the markdown report). Randomness is seeded; every figure in this document is computed from the database at build time.

Data sources: Federal Reserve Economic Data (FRED/ALFRED), Federal Reserve Bank of Cleveland inflation nowcast, and Yahoo Finance. Methods: ridge regression and gradient-boosted trees (LightGBM).